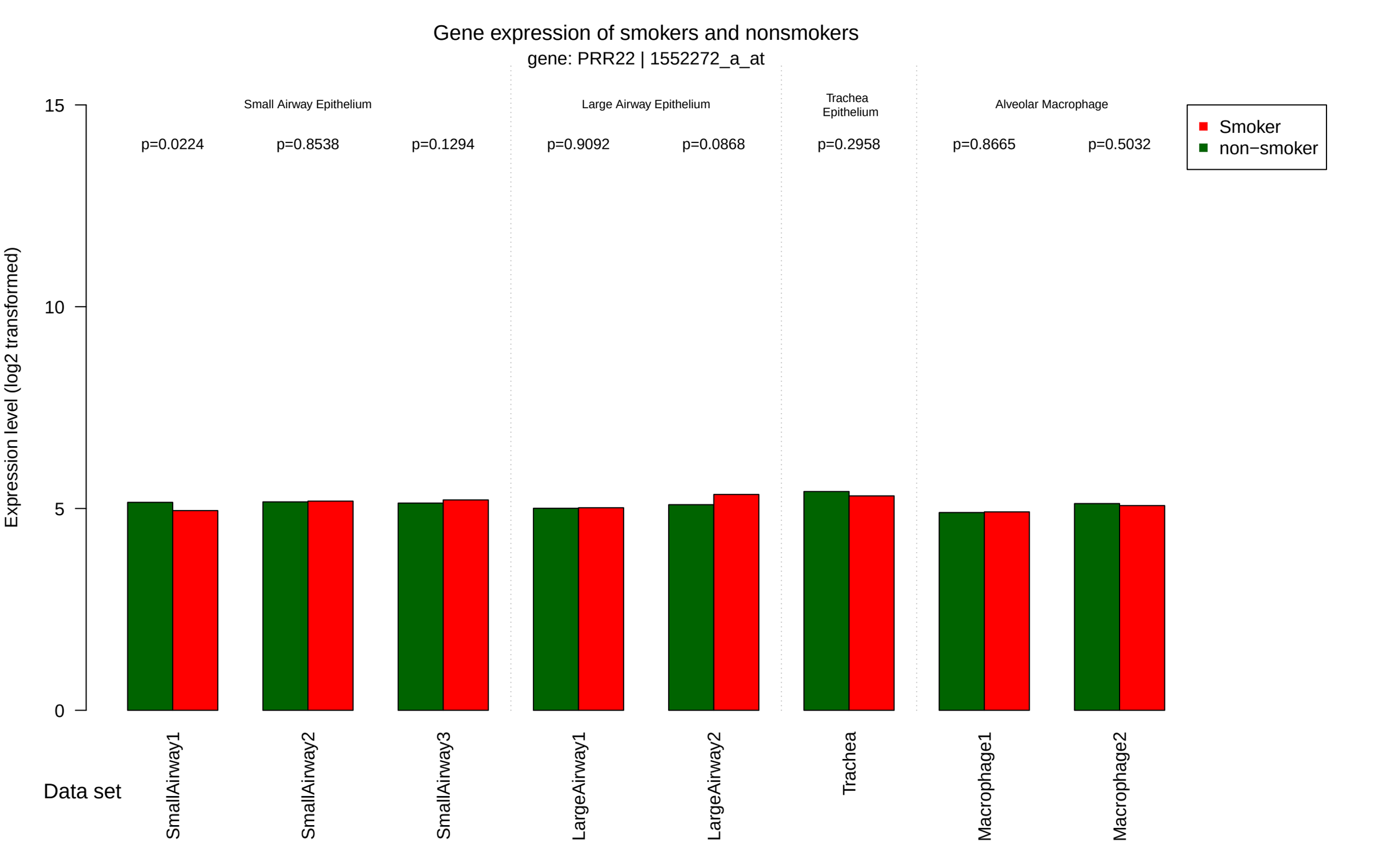
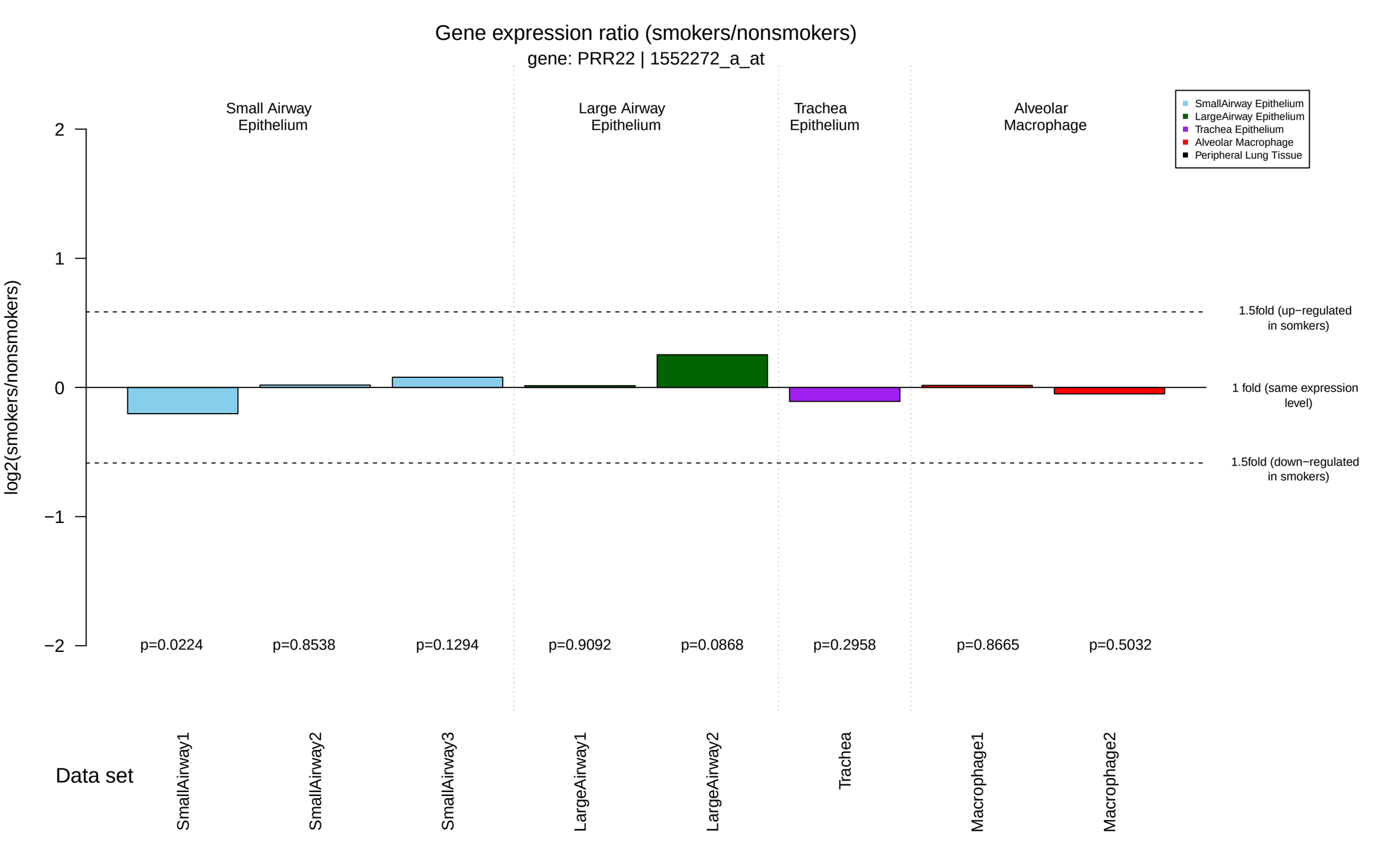
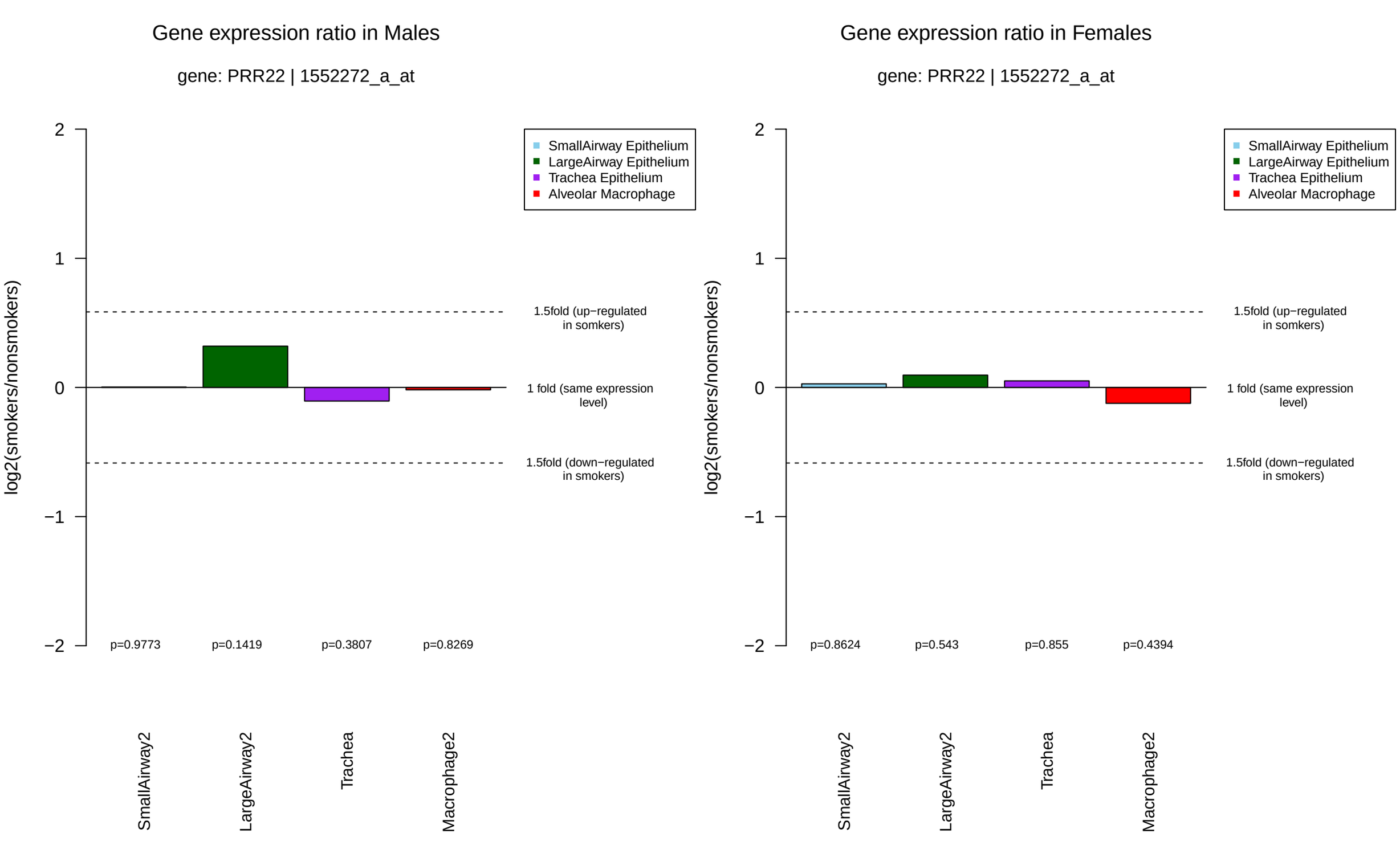
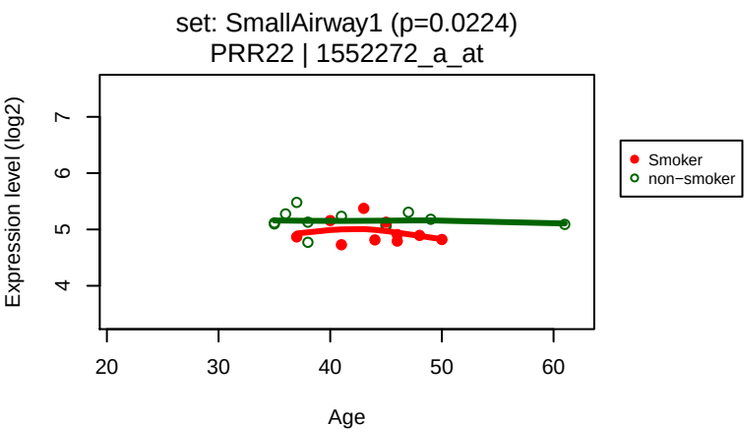
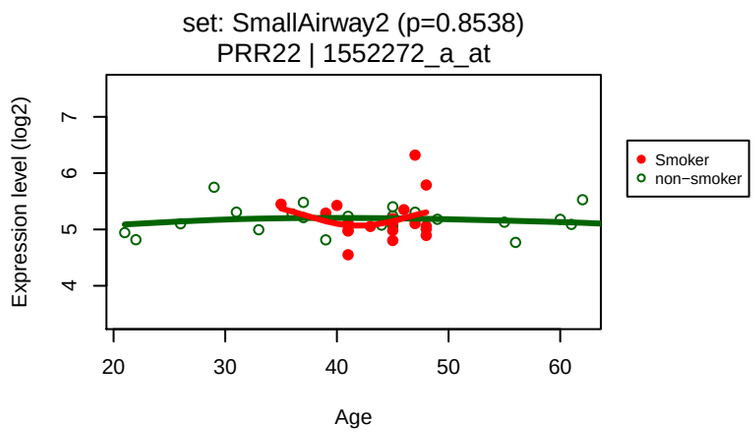
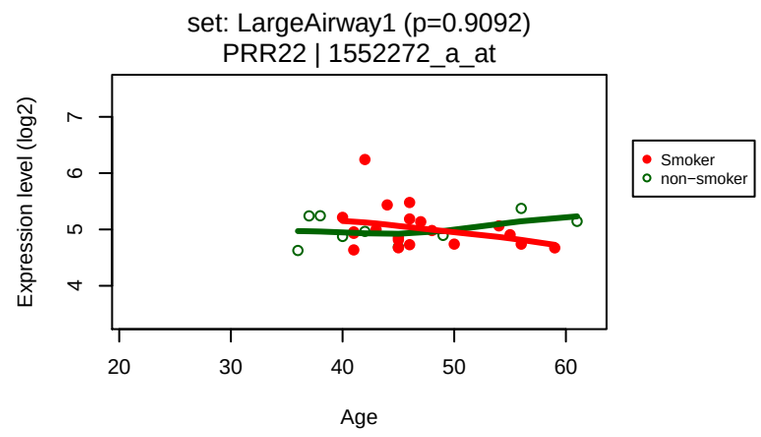

Gene expression of smokers and nonsmokers
gene: PRR22 | 1552272_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0224
p=0.8538
p=0.1294
p=0.9092
p=0.0868
p=0.2958
p=0.8665
p=0.5032
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

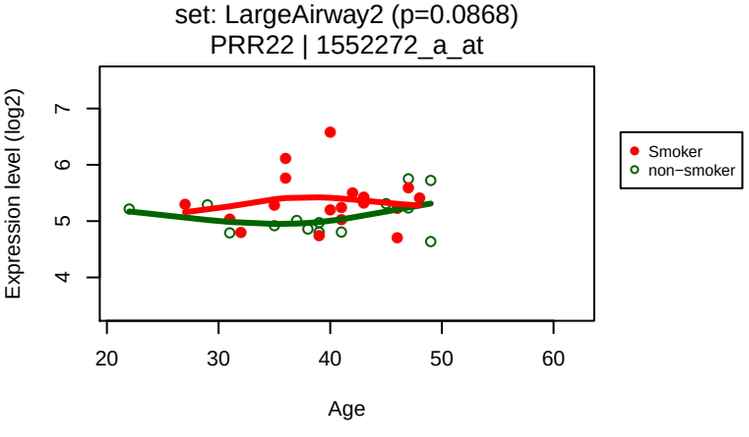

Gene expression ratio (smokers/nonsmokers)
gene: PRR22 | 1552272_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0224
p=0.8538
p=0.1294
p=0.9092
p=0.0868
p=0.2958
p=0.8665
p=0.5032
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

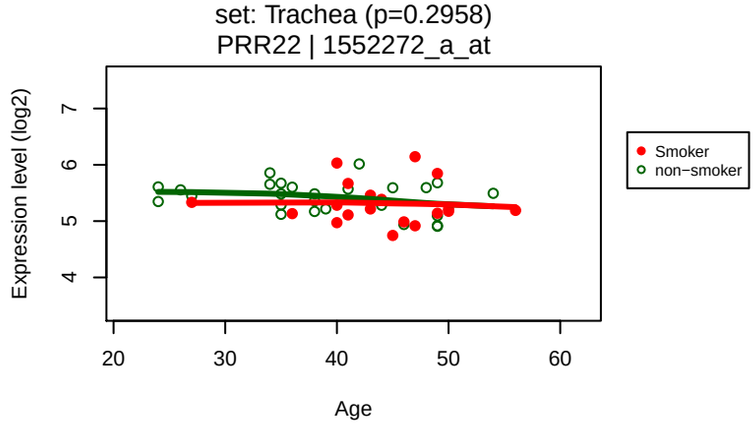

Gene expression ratio in Males
gene: PRR22 | 1552272_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9773
p=0.1419
p=0.3807
p=0.8269
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRR22 | 1552272_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8624
p=0.543
p=0.855
p=0.4394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

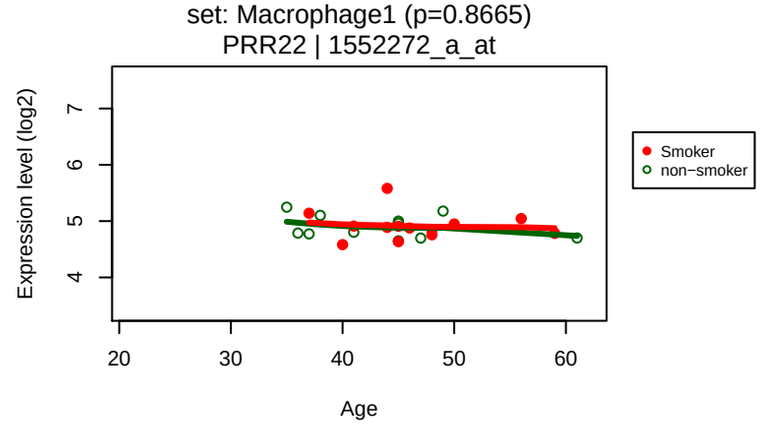

set: SmallAirway1 (p=0.0224)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker

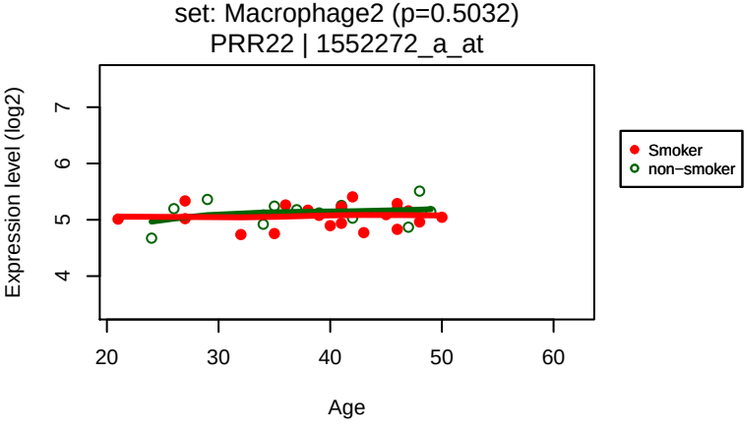

set: SmallAirway2 (p=0.8538)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker

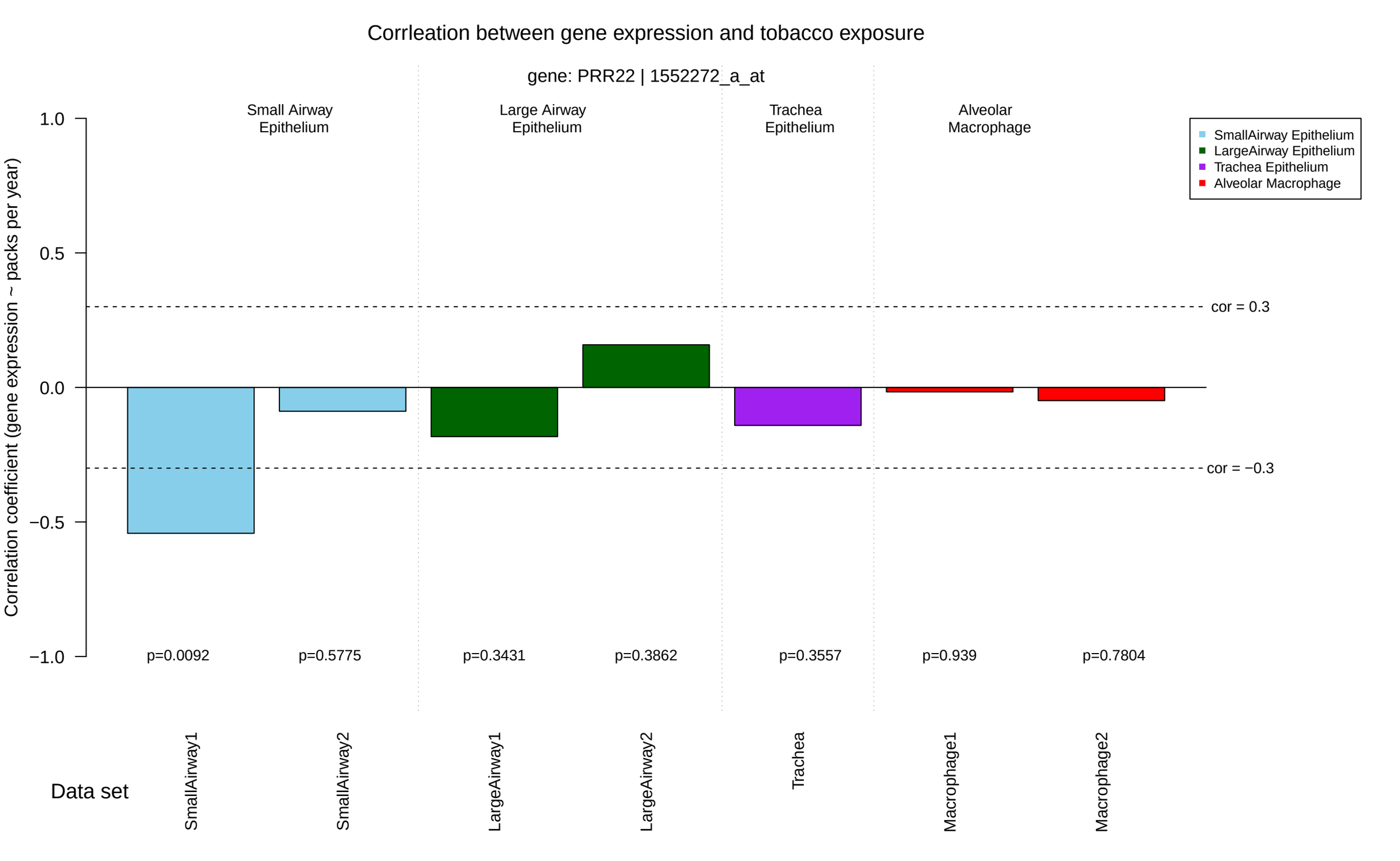

set: LargeAirway1 (p=0.9092)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0868)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2958)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8665)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5032)
PRR22 | 1552272_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRR22 | 1552272_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0092
p=0.5775
p=0.3431
p=0.3862
p=0.3557
p=0.939
p=0.7804
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2